

1. Increased Throughput

Concept: Adding more processors should lead to a higher throughput, but the relationship is not necessarily linear.

Problem:

Suppose you have a system with 1 processor that can process 100 tasks in 10 hours. If you add

N processors, each processor processes tasks independently but with some overhead, leading to a decrease in efficiency. The efficiency can be modeled as:

$$E(N) = \frac{N}{1 + 0.1N}$$

Where $E(N)$ is the efficiency with N processors.

Question:

Calculate the expected time to process 1000 tasks with 1, 2, and 5 processors.

Solution:

- **For 1 Processor:**

- Efficiency $E(1) = \frac{1}{1+0.1 \times 1} = 0.909$
- Time = $\frac{1000 \text{ tasks}}{100 \text{ tasks per hour}} = 10 \text{ hours}$

- **For 2 Processors:**

- Efficiency $E(2) = \frac{2}{1+0.1 \times 2} = 0.833$
- Tasks processed per hour = $2 \times 100 \times 0.833 = 166.6$
- Time = $\frac{1000}{166.6} \approx 6 \text{ hours}$

- **For 5 Processors:**

- Efficiency $E(5) = \frac{5}{1+0.1 \times 5} = 0.714$
- Tasks processed per hour = $5 \times 100 \times 0.714 = 357$
- Time = $\frac{1000}{357} \approx 2.8 \text{ hours}$

2. Economy of Scale

Concept: More processors can lead to cost savings.

Problem:

Consider that a single processor system costs \$500 in total for a CPU, RAM, and storage. A multiprocessor system with N processors can share the cost of peripherals. The cost of the multiprocessor system can be modeled as:

$$C(N) = 500 + 300 \cdot \left(\frac{N}{2}\right)$$

Where 300 is the cost shared for peripherals.

Question:

1. Calculate the total cost for 1, 2, 4, and 8 processors.

Solution:

- **For 1 Processor:**

- $C(1) = 500 + 300 \cdot \left(\frac{1}{2}\right) = 500 + 150 = 650$

- **For 2 Processors:**

- $C(2) = 500 + 300 \cdot \left(\frac{2}{2}\right) = 500 + 300 = 800$

- **For 4 Processors:**

- $C(4) = 500 + 300 \cdot \left(\frac{4}{2}\right) = 500 + 600 = 1100$

- **For 8 Processors:**

- $C(8) = 500 + 300 \cdot \left(\frac{8}{2}\right) = 500 + 1200 = 1700$

3. Increased Reliability

Concept: Distributing functions across processors can increase reliability.

Problem:

Imagine a system where a single processor fails with a probability of p . If you have N identical processors, the probability of failure for the entire system can be modeled as:

$$P(\text{failure}) = p^N$$

Question:

1. If the probability of failure for a single processor is 0.1, calculate the overall failure probability for 1, 2, and 3 processors.

Solution:

Conclusion

These mathematical problems illustrate key concepts in system performance, cost efficiency, and reliability when scaling up processor usage. Students can engage with these problems, analyze the results, and discuss implications, enhancing their understanding of multiprocessing systems.

For 1 Processor:

- $P(\text{failure}) = 0.1^1 = 0.1$

For 2 Processors:

- $P(\text{failure}) = 0.1^2 = 0.01$

For 3 Processors:

- $P(\text{failure}) = 0.1^3 = 0.001$

Lab 1:

1. Increased Throughput

```
def calculate_throughput(num_processors, tasks=1000):
    efficiency = num_processors / (1 + 0.1 * num_processors)
    tasks_per_hour = num_processors * 100 * efficiency
    time_required = tasks / tasks_per_hour
    return time_required
# Test with 1, 2, and 5 processors
for processors in [1, 2, 5]:
    time = calculate_throughput(processors)
    print(f"Time required with {processors} processor(s): {time:.2f} hours")
```

2. Economy of Scale

```
def calculate_cost(num_processors):
    base_cost = 500
    shared_cost = 300 * (num_processors / 2)
    total_cost = base_cost + shared_cost
    return total_cost
# Test with 1, 2, 4, and 8 processors
for processors in [1, 2, 4, 8]:
    cost = calculate_cost(processors)
    print(f"Total cost for {processors} processor(s): ${cost:.2f}")
```

3. Increased Reliability

```
def calculate_failure_probability(num_processors, failure_probability=0.1):
    return failure_probability ** num_processors
# Test with 1, 2, and 3 processors
for processors in [1, 2, 3]:
    failure_prob = calculate_failure_probability(processors)
    print(f"Failure probability with {processors} processor(s): {failure_prob:.4f}")
```